Networking Assignment Scenario – Inter-building Routed Network Design.

Scenario Overview:

Your university campus has **three buildings**, each with a different network design and routing protocol. You are tasked with designing and implementing an inter-building network using **Packet Tracer**, with the following criteria:

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1. Network Layout:

- **Student Building**
  - Uses the **RIP** routing protocol.
  - Contains **3 VLANs**:
    - `MCLA` – for undergraduates
    - `MCLB` – for postgraduates
    - `MCLC` – for research students
  - Each VLAN is assigned its own subnet.
  - A Layer 3 switch handles **inter-VLAN routing**.
- **Admin Building**
  - Uses the **OSPF** routing protocol.
  - Divided into at least **2 subnets**.
  - One subnet is configured to use **dynamic NAT** to access external resources.
  - Can fully communicate with all IT Building subnets.
- **IT Building**
  - Uses **EIGRP** for internal routing.
  - Divided into at least **3 subnets**, including a special subnet named **"Exam"**.
  - Hosts a **dedicated DNS server** for resolving the domain:
    `www.confidential.com` → `192.168.60.10`

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2. Inter-Building Communication:

- To connect all three buildings together, use **routing protocol redistribution** (RIP ↔ OSPF ↔ EIGRP).
- Implement **route redistribution** at the border routers where protocols meet.

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3. Access Control Rules:

- **Admin Building** can communicate with **all subnets** in the IT Building, including "Exam".
- **Student Building** can communicate with all IT subnets **except** the "Exam" subnet.
    - Apply an **ACL** (Access Control List) on the IT router or at the redistribution point to **deny Student building access to the "Exam" subnet only**.

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4. Additional Configuration Requirements:

- Set up **Dynamic NAT** in the Admin building so that internal private IPs (from the NAT subnet) are translated to a pool of public IPs.
- Configure **DNS** on the IT building's DNS server so that clients can resolve:
- `www.confidential.com → 192.168.60.10`
- Test name resolution and connectivity using `ping` and `nslookup`.

Deliverables:

1. **Packet Tracer (.pkt) file** with full implementation.
2. A brief **network topology diagram** showing:
    - Routers, switches, VLANs, and key devices (DNS server, NAT, etc.).
    - IP addressing plan.
3. A **document** (or PDF) including:
    - IP address table.
    - VLAN assignments.
    - Routing protocols and redistribution points.
    - NAT configuration summary.
    - ACL applied and justification.

Bonus (Optional):

- Add **DHCP** servers in each building for dynamic IP allocation.
- Enable **SSH** access to all routers for secure remote management.
